

Lesson 2.10.1: Solving Word Problems Involving Quadratic Functions and Equations

Jonathan R. Bacolod

Solving Quadratic Equations

Quadratic equations may be solved by:

factoring

completing the square

the quadratic formula

using the graph of the function

Zeros or Roots

The **zeros** or **roots** of a quadratic function are the values of x where

$$f(x) = 0.$$

Real-Life Applications

Many **real-life situations** can be modeled using quadratic equations and functions.

area

motion

geometry

business

Interpreting Solutions

Always interpret the answer in the context of the given problem.

Reject **impossible values** when necessary.

For example, a length or time cannot be negative.

Examples

Example 1

The area of a square is 169 cm^2 . Find the length of one side.

Let x be the length of one side.

$$x^2 = 169$$

$$x^2 - 169 = 0$$

$$(x - 13)(x + 13) = 0$$

$$x = 13 \quad \text{or} \quad x = -13$$

Since length cannot be negative,

$$\boxed{x = 13 \text{ cm}}$$

Example 2

A rectangle has a length that is 5 m longer than its width. Its area is 84 m².

Let the width be x m.

Length:

$$x + 5$$

Area:

$$x(x + 5) = 84$$

$$x^2 + 5x - 84 = 0$$

$$(x + 12)(x - 7) = 0$$

$$x = -12 \quad \text{or} \quad x = 7$$

Therefore,

$$\boxed{7 \text{ m by } 12 \text{ m}}$$

Example 3

A ball is thrown upward and its height is

$$h(t) = -5t^2 + 20t + 2.$$

The ball hits the ground when $h(t) = 0$.

$$-5t^2 + 20t + 2 = 0$$

Using the quadratic formula:

$$t = \frac{-20 \pm \sqrt{20^2 - 4(-5)(2)}}{2(-5)}$$

$$t = \frac{-20 \pm \sqrt{440}}{-10}$$

$$t \approx -0.10, \quad 4.10$$

Since time cannot be negative,

$$\boxed{t \approx 4.10 \text{ s}}$$

Practice Exercises

Practice 1

The area of a square is 169 cm^2 . Find the length of one side by writing and solving a quadratic equation.

Practice 2

A rectangle has a length that is 5 m longer than its width. If its area is 84 m^2 , find its dimensions.

Practice 3

A ball is thrown upward and its height after t seconds is

$$h(t) = -5t^2 + 20t + 2.$$

After how many seconds will the ball hit the ground?

Practice 4

The product of two consecutive integers is 156.
Find the integers.

Practice 5

A garden has a length that is 4 m more than its width. Its area is 96 m^2 .

Find its dimensions.

Practice 6

Solve using the quadratic formula.

$$3x^2 - 7x - 6 = 0$$

Practice 7

The profit of a small business is modeled by

$$P(x) = -2x^2 + 40x + 120,$$

where x is the number of items sold (in tens).
Find the maximum profit.

Practice 8

A projectile follows the path

$$h(t) = -4.9t^2 + 29.4t.$$

Determine when the projectile returns to the ground.

Activities

Activity 1

A rectangle has a length that is 7 m longer than its width. If its area is 144 m^2 , find its dimensions.

Activity 2

The product of two consecutive integers is 210.
Find the integers.

Activity 3

Solve using the quadratic formula.

$$2x^2 - 9x - 5 = 0$$

Activity 4

A rectangular picture frame has an outer dimension of 20 cm by 28 cm.

A uniform border of width x cm leaves a picture area of 288 cm^2 .

Find the width of the border.

Activity 5

The sum of two numbers is 20 and their product is 84.

Find the two numbers.

Thank you!